Strip Arithmetic II — Slope, Width & the Singular Series

Wessen Getachew · 2026 · one self-contained `index.html`, no build step, no server.

A browser tool for looking at one object from many angles: the triangle of cells **(r, M)** with 1 ≤ r ≤ M ≤ G. A cell is *coprime* when gcd(r, M) = 1. Every view, overlay and sheet on the page is a different way of reading that triangle, and each rests on a classical result (Euclid, Euler's φ, Möbius inversion, the Chinese remainder theorem, Hardy–Littlewood local factors). **Nothing here is offered as a new theorem.** Exact counts are computed in your browser; anything only observed is labelled as observed (see §4 · Status on the page).

This is Strip II of the C(n) framework. The page has three layers, entered in this order:

Layer	What it is	Range
Intro	A timed animation you can scrub and play with	rings M up to 512
Opener	The welcome panel: the foundational layer, a sieve game, live previews	G up to 1024 (Level 1)
Main page	The full workbench: lattice, strips, comet, status	G up to 10,000

Open `index.html` in any modern browser. On the first visit the intro plays; it is skipped on later visits and whenever the link carries a saved state (`#s=…`).

The object

- **Cell (r, M).** Ring M holds the fractions r/M.
- **Coprime colouring**, density $1/\zeta(2) = 6/\pi^2$ in the limit.
- **Block sieve C(n).** A cell survives to depth *n* when r is coprime to every one of M, M+1, …, M+n. Primes p ≤ n+1 remove every cell with p | r; a prime p > n+1 removes a fraction (n+1)/p². The survivors tend to
  $C(n) = \zeta(2)\cdot\prod_p (1 - \min(n+1, p)/p^2)$, with C(0) = 1, C(1) ≈ 0.5307, C(2) ≈ 0.4128.
- **Strip.** The band s ≤ b·r − a·M < s + w for slope a/b, offset s, width w. Slope +1 with width n+1 is the block-coprime density; slope −1 with width 1 is the Goldbach anti-diagonal r + M = N, tied to the Hardy–Littlewood singular series 𝔖(N).

The views

View	Placement of cell (r, M)
Grid	A square lattice: r across, M up
Ring	Ring M at a radius, cell r at angle θ
Cayley	The ring plane under Γ(z) = (z−1)/(z+1), radius $\rho = \rho_0 + \rho_D\cdot M/G$
Helix	The ring with height θ/2π: the angle is not wrapped, and the tail is visible when you rotate

Angle modes (Ring, Cayley, Helix):

- **clamp** (default): θ = 2π·min(r/M, 1) — r ≥ M is pinned at the limit, so one turn.
- **wrap**: θ = 2π·r/M, unbounded, so r > M laps the ring and draws the full G × G square. On the Helix this is the *tail*.
- **unit**: θ = 2π·r/G, radial spokes budgeted by G, not M.

The angle scale k in θ = kπ·r/M is adjustable (k = 2 is the full turn above). Views are presentation only; no count depends on the view.

Sectors, overlays and colour

- **Sectors** S_s = { ⌊M/r⌋ = s }: the first step of the Euclidean algorithm. The wedge for S_s lies between the rays r/M = 1/(s+1) and 1/s.
- The intro's **sectors** button cycles **4 · 8 · 16 · all** (default 16). Every sector drawn gets its own hue; sectors 1–8 keep the original palette, further ones get spread hues, and anything past the last drawn sector is pooled grey.
- **Sector labels** s=1, 2, … sit on their own wedge and shrink with the room the wedge has (s ≥ 6 are smaller), fading rather than jumping when k squeezes the wedges.
- **Corner tables** show counts for the first sectors one by one (at most 10) with a single pooled row for the rest.
- **Prime spiral** r = p on every ring M ≥ p (a hyperbolic spiral), with its **mirror** r = M − p (complex conjugation); ×/÷ = the multiplicative side.
- **±n path** r = (M ± n)/2, the solutions of 2r ≡ ±n (mod M). **Connect** joins spiral and path at shared rings; lines that are about to be cut at n → n+1 are shown red.
- **Goldbach** (+/−): pairs g + s = E drawn as the four Strip I lines; the page sets E = 2(G+1).

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1 · The Intro

A full-screen animation on a timeline you can scrub; the clock starts at −30 s.

- **Opening (−30 s to −20 s).** Starts at M = 1 and grows to M = 512 while k unwinds from 2 to 0 on the Helix. While M ≤ 7 every point is labelled with its r.
- **Sector rays** draw outward one after another, then settle to faint lines.
- **Time slider** at the top, and a **Restart** mark on the title. **Enter →** goes on to the opener.

Controls (ᵥ *controls*):

Group	What it does
Ring	M slider and ▲ ▼ step, ► **M** (play 1 → 512 and back), **512**, angle mode, k, **hole** (inner radius; at 1 every ring collapses to one circle), view turns **x · y · z** with 0 / ±45 / ±90 presets, ρ_0 and ρ_D, point **size**, **colour** (sector vs. the opener's colouring), **sectors** 4/8/16/all
Prime spiral	pick p, or scan the primes up and back
Goldbach	pair slider and ► **+ −**
Slope band	a/b, offset, width; ► **grow 2w / −1s**; **lock** to the Goldbach pair on screen
r census	counts of r inside the slope band
Sieve C(n)	depth n 0–255, ► **C(n) play**, G sweep

Group	What it does
Record	up to 256 s of every parameter change; **clear**; **video** (MP4 where the browser can, else WebM, 30 fps)
View & export	**reset view**, **fit tail**, **full** screen

Buttons under the plate: ∑ formula (keep the top corner tables, show the equation playing at this moment with live numbers), **math sheet**, **→ opener**, **→ main** (send the exact picture on), **1080×1920** PNG.

Navigation: wheel/pinch to zoom, drag to pan, double-click to reset. **fit tail** sets zoom and pan so the whole helix, tail included, is in the plate while keeping your x/y/z turn.

> Zooming out in Cayley normally adds rings so the new space fills with points; with Goldbach on it does not, so the locked pair/slope stay put.

2 · The Opener

Title bar *The Opener · Strip Arithmetic II*: ♫ **Music** (the track is *Riemann Zeta Function* by Vince Rubinetti; it starts on your first click or key press), 🔗 **Share link** (reopens the opener with the exact same settings), **×** close.

- **Level 1 · Run the sieve.** A game: a cell is crossed out when a prime you pick divides gcd(r, M). Cross out 99 % of the cells that share a factor using as few primes as you can; *par* is the best greedy order. A hint shows the best next prime but caps the stars at 2. **Enter Level 2** goes to the main page. Colours: teal = coprime, dark = crossed out, coral = shares a factor and still in the game.
- **What to know before exploring** — the foundational layer (G lattice in Grid / Ring / Cayley form, colouring, lift dynamics) and how Strip II builds slope a/b · offset s · width w on top of it. **Gap presets** (1, twin 2, cousin 4, sexy 6, 8, 10, 12; ‹ pair / pair ›; all pairs).
- **What this page adds** — the strip definition, the C(n) and ℭ(N) readings, and **presets**: diagonal, block n = 1 / 2 / 4, Goldbach, slope ½ (the default strip).
- **Preview the math sheet** and **Preview the sieve only** open as scrollable pop-ups with the same content the sheet has.
- **Live preview: view** (Grid · Ring · Cayley · Helix · All 3 — one view at a time redraws faster), **hole** and **circle method** (the Farey dissection into major arcs drawn in the hole), **angle** (wrap · clamp · unit), kπ, ⛶ **Lock canvas** (pin the canvases to the top while you scroll the controls), **labels** with ▲ ▼ mode and size (per-point labels, only while G is small), **corner stats** toggles (gcd = 1, sectors, sieve, primes).

3 · The Main page

Slope, Width, and the Singular Series. Five folds:

- **§0 Guide** — the whole thing in classical terms: the three views, the sieve step by step, lift overlays, block pattern and M set, sectors/strips/slopes, Goldbach and the pair finder, the sheets, and a table of *which classical result is behind which control* with its status.
- **§1 Setup** — blocks, strips and the residue count: the rule underneath, two normalisations and where they part company, the ℭ branch (slope −1) and the D branch (the block).
- **§2 The lattice** — the workbench.
  - **Plate:** zoom − / + / Reset, **Lens**, point **size** and **shape** (square, circle, triangle), Cayley **guides**.
  - **G:** number box (2 – 10,000), −/+, **◄P / P►** to step to the previous/next prime, ► **Play**, presets 2 · 16 · 64 · 109 · 256 · 1k · 2k · 4k · 10k, play speed.
  - **Slope a/b** with presets (+1 diagonal, −1 Goldbach, farey rays), **Lock strip to pairs**, **Which r** (kept / removed by sector), **The block** (n = 1, 5, 29 …), **Walk** (strip or band), one-line presets.

- **§3 The comet** — the Goldbach comet r(N) against N, with the same $\mathfrak{C}(N)$ normalisation; N max up to 300,000. Includes a note on the $li_2(N)$ stand-in and why the "with $\mathfrak{C}$" column sits near 0.95–0.97 instead of 1.
- **§4 Status** — provenance of each statement: elementary and classical, one classical theorem, one classical evaluation, one open conjecture, and what the page does *not* claim.

4 · The sheets

Math sheet and **sieve-only sheet** export the state on screen and open as a page you can print to PDF. Contents: parameters, derivations with the exact numbers, plates, and in advanced mode isolated plates per layer. Numbered parts include: the three views of the same cells · legend and definitions · state at this moment · sectors · kept ratio against G (same n) around the target D(n) · prime columns r = p · births φ(M) per ring · D(n) and C(n) = ζ(2)·D(n) · the slope band · Goldbach (E, formula, circle method, actual).

The sheet's parameters follow the live state: view and its turns, zoom and pan, angle mode, colouring, sectors drawn, point size, Goldbach pair, strip a/b · width · offset, and the C(n) lock. The Helix and Cayley plates keep the zoom and pan you set.

5 · Sharing and saving

- **Share link** — the page URL plus `#s=<state>`; only what differs from the defaults is stored.
- **→ opener / → main** from the intro hand the exact picture (G, n, view, angle, hole, tilt) on.
- **Intro recording** — up to 256 s, replayable and exportable as video.
- **PNG** — 1080 × 1920 from the intro; sheet plates from the sheets.
- **Seen-it flag** — the intro skips itself after the first visit (browser `localStorage`, key `stripII_welcome_seen`).

6 · Limits worth knowing

- The intro draws up to 512 rings. The opener's sieve game goes to 1024. The main page goes to G = 10,000.
- In *wrap* and *unit* modes with a large G the Helix tail is very tall (about G turns), so **fit tail** will shrink the main ring a lot.
- With `sectors: all` and a large G there are more sectors than distinguishable hues, so some colours repeat.
- Per-point labels only appear while G is small.

Credits and status

Everything is classical material restated and computed in the browser: Euclid, Euler, Bézout, Möbius inversion, Eratosthenes–Legendre sieve, CRT, Mertens' theorems, Hardy–Littlewood local factors. The most accurate name for the object is an *Euler-product block sieve on the coprime lattice*. Full attribution, with chips that open each result on the page, is in the credits panel at the bottom of the main page. The music is *Riemann Zeta Function* by Vince Rubinetti.